SOUTH LAKE HOME EDUCATORS



Lesson 7

Core Mechanics I: On the Move

Semester 1, Week 7 - Core Mechanics Unit Begins

At a Glance

Lesson 7 objectives & materials

By the end of today, students will:

- Implement continuous, frame-rate-independent movement
- Distinguish "destroyed by a hit" from "destroyed by escaping"
- Explain why generic "on destroyed" hooks can silently misfire
- Relocate state-changing logic to the specific event it belongs to

What We're Using Today

- Demo laptop with all 8 engines verified
- Godot, for the live lecture demo
- GameMaker, for the bug-discussion contrast
- No new assets this week - logic-only lesson
- Your own [YourName]Game project from Lesson 6 - not a new one

IGDA framework tie-in: Programming - game loops and continuous simulation, event-driven programming, debugging and code correctness (introductory).

Today's Agenda



0:00 - 0:05	Welcome back, Lesson 6 recap, milestone check-in
0:05 - 0:20	New Unit: Core Mechanics - what's ahead through Lesson 11
0:20 - 0:40	Movement That Doesn't Stop: Continuous Motion
0:40 - 1:00	The Hidden Bug: When "Destroyed" Doesn't Mean "Defeated"
1:00 - 1:05	Introduce lab exercise, hand out engine-specific handouts
LAB TIME	***— On the Move —***
1:05 - 1:55	Advance enemies toward the player + escape cleanup
1:55 - 2:00	Exit check / quick share-out



Welcome Back

Lesson 6 recap - you hit the Level 1 MVP milestone.

5 min

Welcome Back

5 min

Quick recap of Lesson 6:

- A working win condition - the final piece of the Level 1 MVP milestone
- A complete, playable game loop: move, shoot, face enemies, win
- Today: enemies stop standing still - the first step into Level 2

Open your own project from last week today, not a new one - and no new assets needed either, this is a logic-only lesson.



New Unit: Core Mechanics

Collision, scoring, win/lose states, simple UI - Lessons 7 through 11.

15 min

Core Mechanics

15 min

You just finished "Programming Basics" - variables, loops, collision, state tracking. Starting today, a new unit builds toward a complete, replayable arcade shooter.

Lesson 7	Enemies advance toward the player (today)
Lesson 8	Lose condition + lives system
Lesson 9	Score tracking + basic on-screen UI
Lesson 10	Game-over and restart flow
Lesson 11	Multiple waves - Level 2 Complete milestone



Movement That Doesn't Stop

Enemies that move every frame, with nothing triggering it.

20 min

Continuous Motion

20 min

So far, everything has moved in response to something - a key press, a spawn. Today's movement just keeps happening, every single frame, forever.

Multiply movement by delta (time since the last frame) to keep speed the same no matter how fast the game runs - the same pattern player movement has used since Lesson 2.

Every enemy also gets a bottom-of-screen check: reach the bottom unshot, and it's removed - but does NOT count as a kill.

Discussion: what else in a game has to run continuously rather than in response to an event? (A countdown timer, a camera following the player, gravity.)



The Hidden Bug

When "destroyed" doesn't mean "defeated."

20 min

Two Ways to Disappear

part 1

Now that enemies move, there are TWO reasons one might stop existing - and only one of them should count as a win.

Shot

Destroyed by a bullet - this IS a kill. Should count toward the win condition.

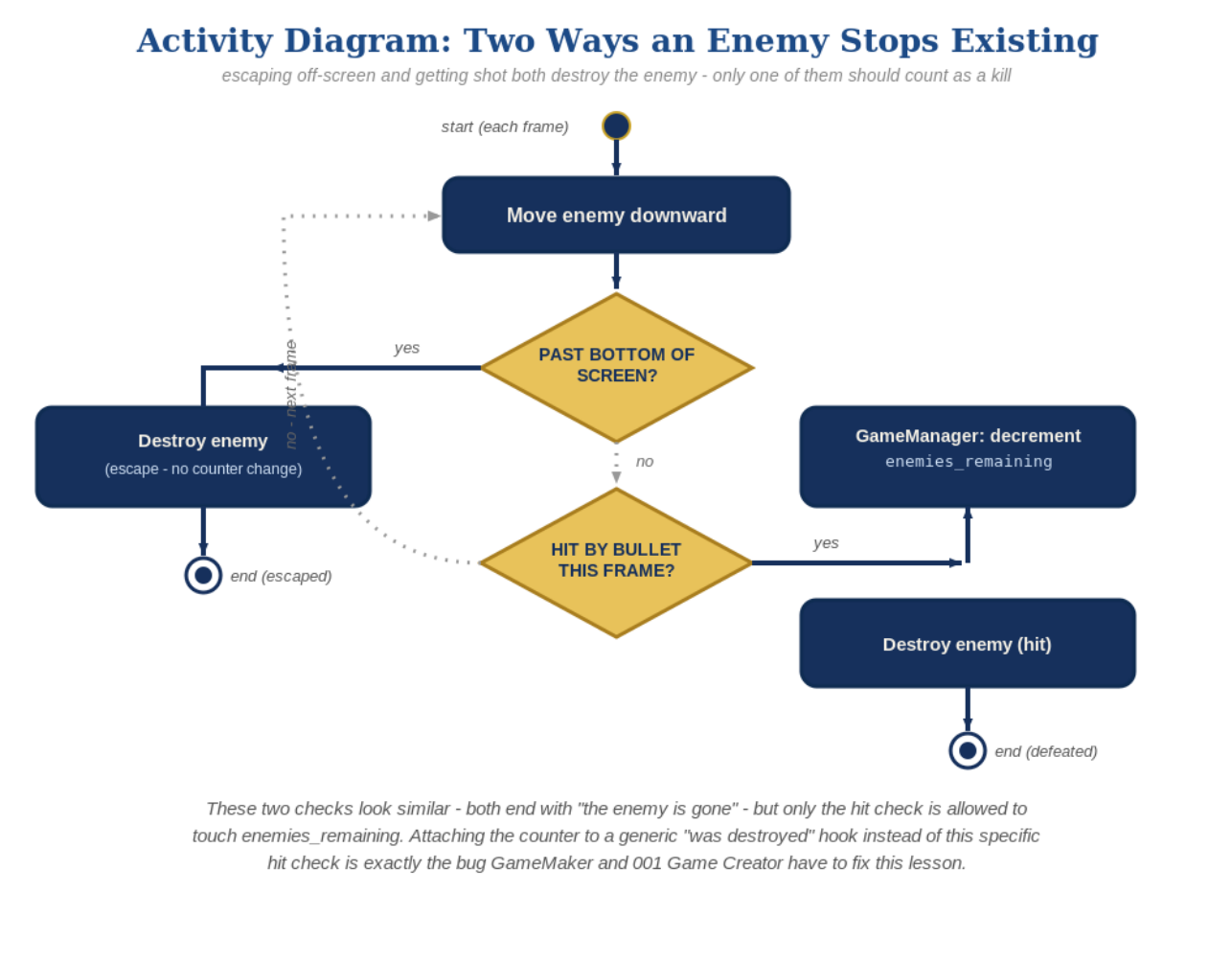
Escaped

Reached the bottom of the screen unshot - NOT a kill. Must never count.



GameMaker's Destroy event and 001 Game Creator's "Defeated / Destroyed" trigger both fire for EITHER reason. Last week's win-counter wiring was correct then - only one destruction path existed. This week it silently breaks.

Two Ways to Disappear, Side by Side



The same shape in every engine - only the hit check is allowed to touch enemies_remaining.

The Fix

part 2

Attach win-counting to the SPECIFIC event that means "got shot," not the GENERIC event that means "stopped existing."

GameMaker	Move the decrement from the Destroy event into the Bullet's Collision-with-Enemy event.
001 Game Creator	Move the decrement from "Defeated / Destroyed" into the existing "Collided with actor" trigger.
Other 6 engines	Already tied the win-counter to the specific hit-detection code in Lesson 6 - nothing to fix.

General principle: attach state-changing logic to the specific event that represents it, not to a generic lifecycle hook that fires for unrelated reasons too.



Today's Exercise & Lab

On the Move - continuous descent and escape cleanup.

10 min

Before We Start: Meet the GameManager

quick aside

One more thing before today's lab - a change already sitting in your Lesson 6 project, not something new to build today.

Your win-counter (enemies_remaining) and win-flag (has_won) now live on their own small dedicated object in every engine - a GameManager - instead of scattered across whichever object happened to need them first.

Scratch	A hidden GameManager sprite, reached by broadcast/receive
Godot	A GameManager child Node, reached by signal connection
GameMaker	obj_gamemanager, a placed instance
Unreal	BP_GameManager, a placed Actor
Unity	Already worked this way - nothing changed for you

If your own project isn't retrofitted yet, do that first today - your updated Lesson 6 handout shows exactly what it should look like.

Why a Dedicated Manager Object?

quick aside

This is a genuinely common real-game pattern, not just a course convention - you'll also hear it called a Game Manager, a Singleton, or Game State.

One clear place to look

Shared, game-wide information has an obvious home - no hunting across multiple objects to find where a value actually lives.

Nothing has to reach into anything else

Any object that needs shared state just talks to the manager, instead of one object reaching directly into another's private variables.

Room to grow

Lives, score, level number, and anything else shared game-wide all have an obvious place to land, starting now - not something to retrofit again later.

Discussion: as we add lives (Lesson 8) and a score (Lesson 9), where should those live? Same answer both times - on the GameManager, right alongside enemies_remaining.

On the Move

10 min

Every enemy advances toward the player, and any that escape are cleaned up without counting as a kill.

Level 1 - Minimal

Enemies move down; an escaped enemy is removed without counting as a kill.

MVP - a complete finish for today.

Level 2 - Complete

A moving enemy can still be shot correctly; escape never triggers "You Win!"

Everything works end to end.

Level 3 - Refined

Varied enemy speeds, and/or a cleaner separation of movement vs. hit logic.

Ready to share.



GameMaker and 001 Game Creator: read your handout's setup section before touching anything - you're deleting code today, not just adding it.

Lab Time Notes

- Most important check this week: let an enemy escape unshot and confirm "You Win!" does NOT appear - for every engine, not just the two with a fix.
- GameMaker / 001 Game Creator: watch for double-counting - the win message triggering after only half the enemies are shot means the old hook wasn't fully removed.
- Enemies never disappearing at the bottom usually means the off-screen threshold doesn't match the project's screen size - adjust the number.
- GameMaker and 001 Game Creator clusters need the most attention this week - circulate there first.

Enemies Are Moving!

Continuous motion, escape cleanup, and a real fix to a latent bug - solid engineering, not just new features.

Next week: a lose condition and a lives system - the first threat to your player.

Try a Level 3 idea at home if you'd like - varied enemy speeds are a fun one to tune.